Even the economics are sobering: “Overall, our lab costs 12-15 Crores per annum to just keep the lights on. The fab will have an opex of ~8000 Crores (500x).”

In 2008, IIT Bombay and IISc launched INUP – the Indian Nano-electronics Users Programme. Ganguly is unequivocal about what it solved. “It created a deep talent pipeline where researchers who had no chance… of creating and sustaining an integrated semiconductor R&D facility with 100+ tools and 500+ processes, had access to such a facility (all expense paid). This was the first green shoots of a national talent ecosystem.”

The outputs are anything but simple: prototypes, patents, publications. Hundreds of PhDs across institutions. And one statistic that quietly delights him: a 40:60 gender ratio. In a sector often accused of gatekeeping, that’s not a small detail.

IITB’s two-decade arc with Applied Materials reads like a case study in how industry-academia partnerships should work.

The donated tools and on-site support unlocked specific capabilities: “Gate Stack Engineering for sub-1 nm,” “Low-resistance metal semiconductor contacts,” “Combinatorial sputtering for memories,” “Atomic Layer etching,” and even “Modeling of 2nm process technologies.”

The co-location mattered. “It enabled training programs like India’s first semiconductor certification program in 2012,” Ganguly notes. Today, Applied sends its New College Graduate hires to train at IITB Nanofab. The relationship spans funded research, consultancy, lab usage, internships, and hiring pipelines.

If he had to distill a playbook for industry partnerships, Ganguly offers three rules: a long-term umbrella agreement aligned at leadership level; ground-level execution with “clear understanding and respect for individual & shared IP”; and building broader consortia instead of one-to-one silos.

India’s semiconductor policy conversation often oscillates between aspiration and anxiety. But Ganguly brings the focus back to physics and capital expenditure, essential building blocks that can't be ignored.

“ATMP (Assembly, Testing, Marking, and Packaging) is 10x smaller investment and operations/execution-focussed with lower technology access risk compared to fab,” he says. Hence, a rational starting point.

Fabs, on the other hand, are “a ‘running on the treadmill’ approach.” First, imported technology transfer. Then relentless indigenous innovation – “constant effort not to fall off the treadmill.”

And that innovation, he argues, comes from “graduate level education with hands-on expertise provided by fab labs – High-end talent – Innovation from fundamental science and engineering.” This is where academia plugs into national strategy – not as spectator sport, but training ground.

If there’s one structural gap India must close, Ganguly points to a pre-competitive semiconductor research center on the scale of imec in Belgium or Albany NanoTech in New York.

“We need an India Semiconductor Research Centre… that translates academic research into industry solutions through an industry consortium model.” A MeitY committee – of which he was part – recommended a $2.5 billion center. “This needs to be accelerated.” It’s the bridge between thesis and throughput, where perspective is a powerful antidote to hype.

Design vs manufacturing

In comparing IIT Madras’ processor sovereignty narrative with IIT Bombay’s manufacturing sovereignty push, Ganguly is careful. “India has had a vigorous chip design ecosystem, though led by MNCs. In comparison, the Indian technology space is barely kick-starting.”

Manufacturing is more consolidated globally, meaning access is harder. “Thus, manufacturing is the primary national technology access challenge that ISM 1.0 chooses to address.”

His answer to the design-only celebration till now? Global relevance. Industry affiliates. And now SemiX – IIT Bombay’s Centre for Semiconductor Technologies – with 12+ industry members and counting.

On indigenous Indian chips over the next 3–5 years, Ganguly is pragmatic. DLI 1.0 showed promise but funding and commercialization remain hard. DLI 2.0’s pari passu model may help, but as Ganguly highlights “designing a chip and finding customers to revenue is a high risk high reward process.”

Another path? Strategic acquisitions of global firms by Indian entities – inheriting customer books instead of starting from zero, Ganguly suggests. Not the stuff of chest-thumping headlines. It’s more patient than that, and perhaps that’s the point.

In a country that loves celebrating design wins and unicorn valuations, Udayan Ganguly is playing a slower, more disciplined game – one that begins with cleanrooms, continues through talent pipelines, and ends, if all goes well, with something far more durable than a press release will ever capture: semiconductor manufacturing sovereignty, built one process step at a time. d